Work Package 4: AI & Recommendation Design - Mohana

Scope:

Design the *intelligence layer*, **not implement it**, perform the market research, state of the arts

Deliverables:

1. Define recommendation approaches:
   - Rule-based filtering

   ⇒ using if-then logic between the words used by the user to recommend books

   - Content-based filtering

   ⇒using NLP to analyse the features of books and using vectorisation to find most suitable book or the closest vector to the given set of features

     1. Document to show various recommendation engines for Books available in research or application
        a. What all apps

   - LLM prompt-based recommendation

   ⇒ Prompt-based recommendation is possible if we use some of the models like Llama3 or others to interact with the user.

2. Define data required:
   - Book metadata

   ⇒ Genres, Author, Summary in under 200 words , Release date, Age restriction.

   - Age filters

   ⇒ We use the age restriction from the book metadata and the account type to make age filters possible.

- Reading history

⇒ We use the books that are already read to suggest new books based on the genres and summary.

3. Write sample prompts for AI

⇒ prince and princess story

⇒ new stories for me

⇒ interesting robot future story

⇒ magic stories for me

⇒ vishnu sharma new stories

⇒ stories which my dad like ??

1. Define how AI integrates with backend:

   - API call flow

   ⇒ Mobile app → Python backend (Chat history)+ MCP Server → LLM Model →MCP Server starts query → RAG vectorises keywords and outputs→ MCP server gives the output details to LLM model → Model makes the output into user-friendly language → Convo completed.

   - Expected JSON structure

   ⇒ Present conversation history to the backend of the app so that the model can understand what are the requests and responses until now.

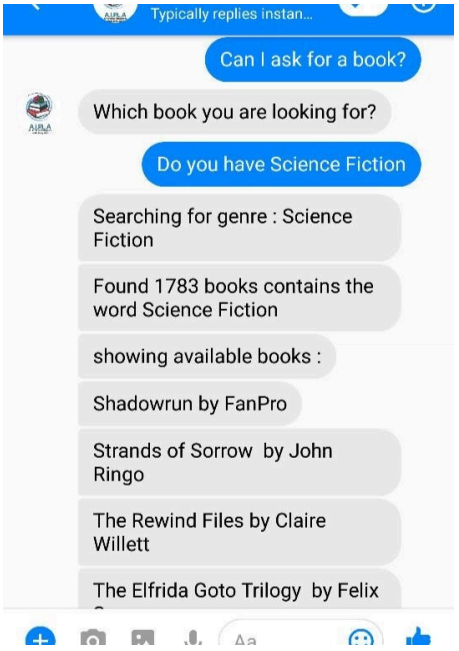
2. Define fallback if AI fails

`Presently analyzing the below repos :`

⇒ Github repos

1. Book recommendation bot ⇒ Can be upgraded RAG based implementation ,presently
   finds similar books based on genre and description similarity using TF-IDF and cosine similarity.

2. Rule-based implementation ⇒ Typical rule-based implementation based on intents.json ,handles only pre-trained data (finding a word and based on that word handle the prompt) and cannot be used in current generation



3. Associative-Rule based learning ⇒ Based on previous books read and borrowed we can suggest some new books like it which are not read (**Get recommendations button**)
4. Complete-Chatbot-Template ⇒ Complete RAG based implementation for the chatbot under MIT license (can be used directly)